

Jigar Kadival

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Objective

I am a dedicated Electronics & Communication Engineering student specializing in firmware development and hardware-software integration. I want to use my problem-solving skills to build robust, innovative embedded system projects, from low-level driver configuration to designing smart, connected IoT interfaces.

Skills

- **Core EC Skills:** Microcontroller Architecture (ESP32-S3, Arduino), Firmware Development, Multi-Resolution Display Interfacing (7-inch, 480x480, 320x480 TFT), Embedded UI Design (LVGL), Memory Management (LittleFS), Hardware-Software Interfacing.
- **Programming:** Embedded C, C++, HTML, Python (basics), Assembly (8051/AVR/ARM - basics).
- **Tools & Technologies:** Visual Studio Code, Arduino IDE, ESP IDF.
- **IoT & Protocols:** MQTT, Wi-Fi, UART, SPI, I2C, API Integration, JSON Data Parsing.
- **Soft Skills:** Problem Solving, Teamwork, Communication, Time Management

Education

College

2022 – 2026

- B.E. in Electronics and Communication
Government Engineering College, Bhavnagar

Embedded Systems Intern

Bonrix Embedded Innovations

Jan 2026 – April 2026

Ahmedabad, Gujarat

- Developed firmware in C for a 7-inch ESP32-S3 HMI smart display, managing JSON data parsing to render dynamic billing interfaces for retail point-of-sale systems.
- Built a smart touch-panel feedback collection UI utilizing LVGL and LittleFS for efficient image asset storage, integrating APIs for real-time data synchronization.
- Engineered an SMS receiver system using a SIM7672S 4G module on a Bharat Pi board, optimizing power management and remote IoT connectivity.
- Engineered custom firmware using the ESP-IDF framework to drive responsive UI layouts across diverse screen form factors, including 480x480 VIEWE smart displays and standard 320x480 TFT modules.
- Developed graphical interfaces for miniature footprint screens (such as compact USB dongle displays), optimizing rendering logic to ensure clear data visualization on constrained hardware.
- Implemented lightweight MQTT messaging protocols on ESP32 hardware to establish low-latency, reliable, and bi-directional communication between remote IoT nodes and central servers.

Projects

Smart Door Lock System

- Designed and implemented an embedded system for a smart door lock using microcontroller (Arduino) and sensors.
- Integrated RFID module, LCD display with I2C interface, and servo motor for secure and user-friendly access control.
- Programmed using embedded C for real-time interaction between RFID reader, LCD status display, and servo motor actuation.
- Applied core concepts of digital electronics, microcontroller interfacing, and hardware-software integration typical in VLSI and embedded systems projects.

Temperature & Humidity Monitoring System using ESP32 (in WOKWI simulator)

- Made a system that measures temperature and humidity using an ESP32 microcontroller and a DHT22 sensor.
- Used a 16x2 LCD with I2C to show live sensor readings on the display.
- Wrote the program in Embedded C/Arduino to read data from the sensor and update it on the LCD.
- Learned and applied microcontroller basics, sensor connections, and hardware–software communication.
- This project can be used for smart home systems and weather monitoring purposes.

Smart IoT Feedback Collection System (ESP32-S3)

- Developed a touch-based smart feedback kiosk using an ESP32-S3 and the LVGL graphics library, featuring emoji-based ratings, dynamic tags, and an on-screen keyboard.
- Engineered a robust Wi-Fi management system and API integration module using HTTPS to sync real-time device mapping and upload user feedback to a custom backend and Firebase.
- Implemented a custom Virtual RAM File System utilizing PSRAM to efficiently cache downloaded image assets, while using LittleFS for offline feedback logging and device configuration.
- Designed a secure, PIN-protected settings interface and an embedded local web server for secure device administration and troubleshooting.